

Project Documentation

1. Introduction

- **Project Title:** CookBook - Your Virtual Kitchen Assistant
- **Team ID:** NM2025TMID44306
- **Team Leader:**

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2. Project Overview

- **Purpose:**
CookBook is a smart web application designed to assist users in cooking by providing personalized recipes, ingredient suggestions, voice-guided instructions, and meal planning support.
- **Features:**
 - Recipe search by ingredients and cuisine
 - Step-by-step cooking instructions with voice assistant
 - Personalized recommendations based on preferences
 - Grocery list generator
 - Nutrition value calculator
 - User login & favorites system

3. Architecture

- **Frontend:** *React.js with Tailwind CSS / Bootstrap*
- **Backend:** *Node.js and Express.js for API and server logic*
- **Database:** *MongoDB to store recipes, users, preferences, and favorites*

4. Setup Instructions

- **Prerequisites:**
 - o *Node.js*
 - o *MongoDB*
 - o *Git*
 - o *React.js*
 - o *Express.js*
 - o *Visual Studio Code*
- **Installation Steps:**

```
# Clone the repository
git clone <repository-link>
```

```
# Install client dependencies
cd client
npm install
```

```
# Install server dependencies
cd ../server
npm install
```

5. Folder Structure

```
CookBook/
|-- client/           # React frontend
|   |-- components/
|   |-- pages/
|
|-- server/           # Node.js backend
|   |-- routes/
|   |-- models/
|   |-- controllers/
```

6. Running the Application

- **Frontend:**

```
cd client
npm start
```

- **Backend:**

```
cd server
npm start
```

- **Access:** Visit `http://localhost:3000`

7. API Documentation

- **User:**
 - `POST /api/user/register`
 - `POST /api/user/login`
- **Recipes:**
 - `GET /api/recipes/search`
 - `POST /api/recipes/add`
 - `GET /api/recipes/:id`
- **Favorites:**
 - `POST /api/favorites/add`
 - `GET /api/favorites/:userId`

8. Authentication

- *JWT-based authentication for secure login*
- *Middleware for protecting private routes*

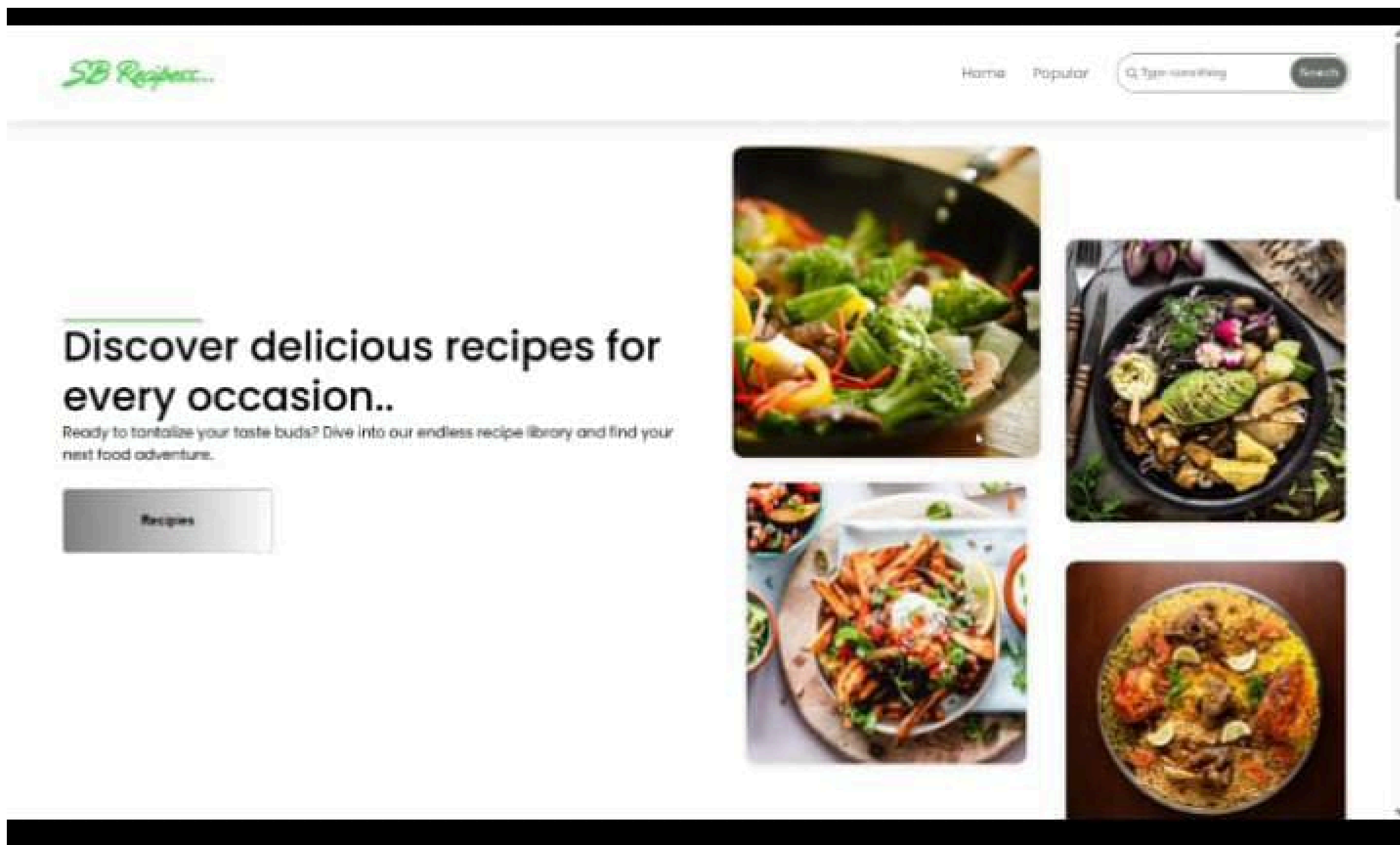
9. User Interface

- *Landing Page with search bar*
- *Recipe Details Page*
- *Step-by-step Cooking Assistant Page*
- *User Dashboard with saved recipes*
- *Grocery List Page*

10. Testing

- **Manual testing:** during milestones
- **Tools:** Postman, Chrome Dev Tools, Jest for unit testing

11. Screenshots or Demo



12. Known Issues

- *Limited voice assistant support on some browsers*
- *Few international recipes not yet included*

13. Future Enhancements

- *AI-powered recipe recommendation system*
- *Multi-language support for instructions*
- *Integration with smart kitchen devices*
- *Offline mode for saved recipes*